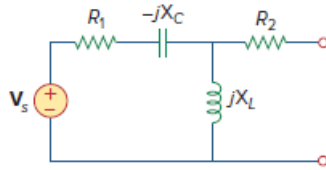


Problem

Using Fig. 10.100, design a problem to help other students better understand Thevenin and Norton equivalent circuits.

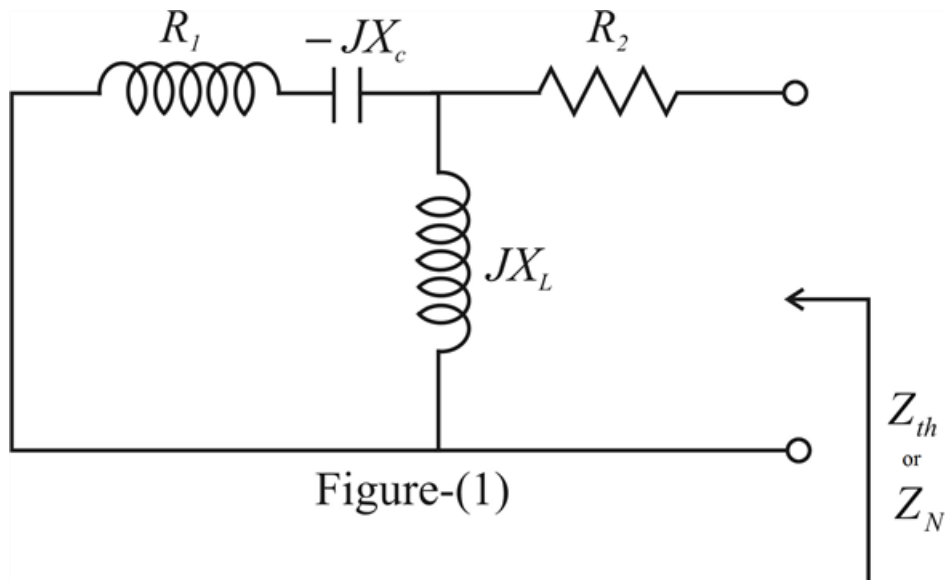
Figure 10.100:



Step-by-step solution

Step 1 of 7

We find Z_{Th} or Z_N by setting the voltage source to zero. As shown in below circuit,



Step 2 of 7

From the above circuit,

$$Z_{Th} \text{ or } Z_N = R_2 + [jX_L \parallel (R_1 - jX_C)]$$

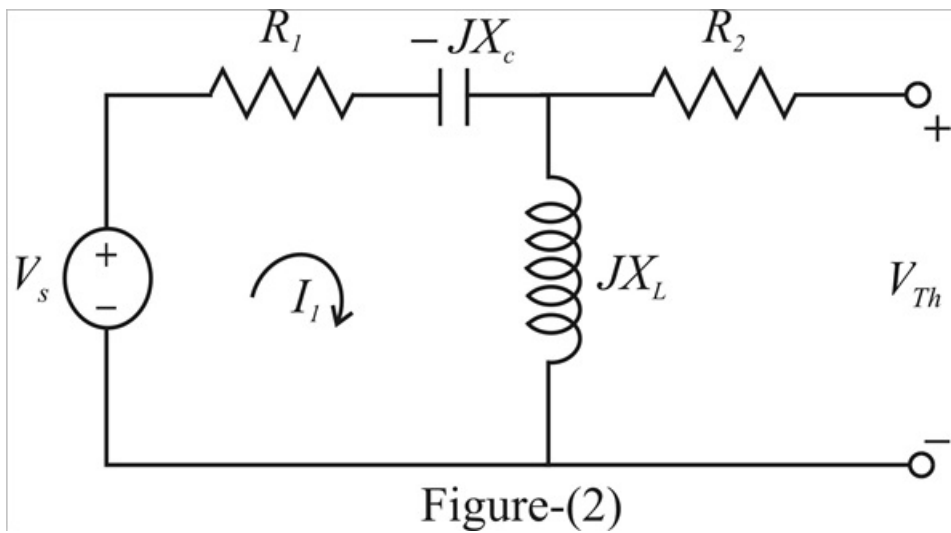
$$Z_{Th} \text{ or } Z_N = R_2 + \left[\frac{jX_L (R_1 - jX_C)}{jX_L + R_1 - jX_C} \right]$$

$$Z_{Th} \text{ or } Z_N = R_2 + \frac{jX_L R_1 + X_L X_C}{R_1 - j(X_L - X_C)}$$

$$\therefore Z_{Th} \text{ or } Z_N = \frac{R_1 R_2 + j(X_L - X_C) R_2 + jX_L R_1 + X_L X_C}{R_1 + j(X_L X_C)}$$

Step 3 of 7

To find V_{Th} consider below circuit,



Step 4 of 7

Applying KVL to above circuit, $-V_s + (R_1 - jX_c)I_1 + jX_L I_1 = 0$

$$I_1 (R_1 - jX_c + jX_L) = V_s$$

$$I_1 (R_1 + j(X_L - X_c)) = V_s$$

$$\therefore I_1 = \frac{V_s}{R_1 + j(X_L - X_c)}$$

From figure -(2)

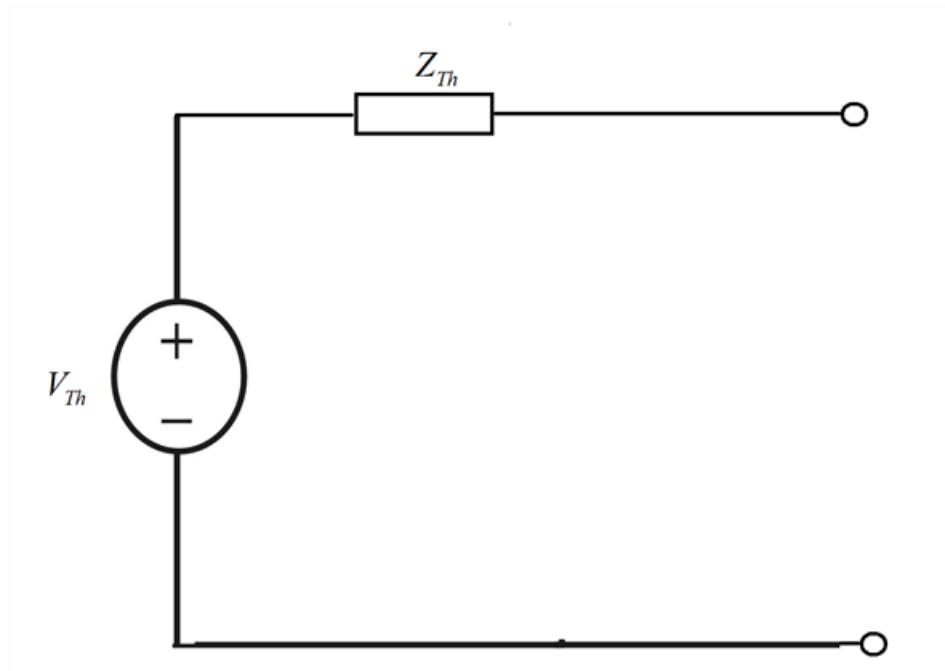
$$V_{Th} = I_1 \times jX_L$$

Substitute I_1 in the above equation.

$$\therefore V_{Th} = \left(\frac{V_s}{R_1 + j(X_L - X_c)} \right) (jX_L)$$

Step 5 of 7

So thevenin equivalent circuit is



Step 6 of 7

Therefore

$$I_N = \frac{V_{Th}}{Z_{Th} \text{ or } Z_N}$$

$$I_N = \frac{\left(\frac{V_s}{R_1 + j(X_L - X_c)} \right) (jX_L)}{\frac{R_1 R_2 + j(X_L - X_c) R_2 + jX_L R_1 + X_L X_c}{R_1 + j(X_L - X_c)}}$$

$$\therefore I_N = \frac{V_s (jX_L)}{R_1 R_2 + j(X_L - X_c) R_2 + jX_L R_1 + X_L X_c}$$

Step 7 of 7

